

Fraud Detection Using Machine Learning

Detecting fraudulent credit card transactions using classification models

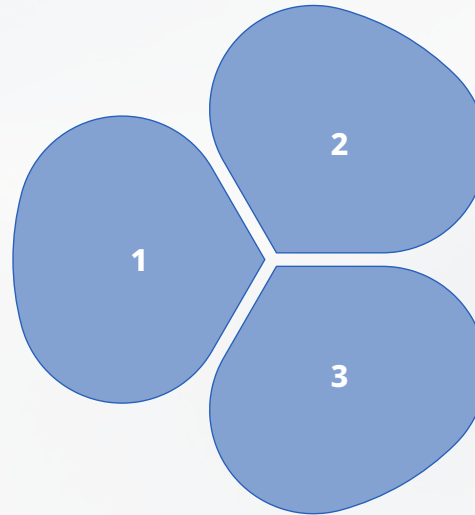
Tools Used: Python, Pandas, Scikit-learn, SMOTE, Seaborn, Matplotlib, Jupyter Notebook

PROJECT OBJECTIVE

Business Problem & Goal

Business Problem

Banks process millions of digital transactions every day. Fraudulent transactions are rare but cause significant financial losses if not detected quickly.

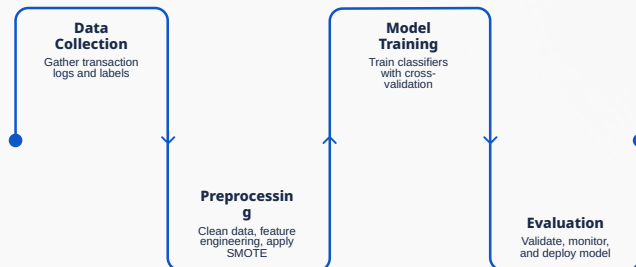


Objective

Build a machine learning model that accurately classifies transactions as **Genuine** or **Fraudulent**.

Expected Outcome

A reliable fraud detection system with high **Recall** and strong overall performance.



Credit Card Fraud Detection Dataset

The dataset contains **284,807 transactions** and **31 features** collected from European credit card transactions.

Feature Descriptions

Feature	Description
Time	Seconds elapsed from first transaction
Amount	Transaction amount
V1–V28	Anonymized numerical features (PCA)
Class	Target: 0 = Genuine, 1 = Fraud

Observation: The dataset contains no missing values, making preprocessing more efficient.

Initial Inspection Performed

01

Load Dataset

Loaded dataset using Pandas

03

Data Types

Verified all feature data types

02

Check Shape

Verified shape and dimensions

04

Missing Values

Confirmed no null entries exist

05

Preview Records

Displayed first five rows of data

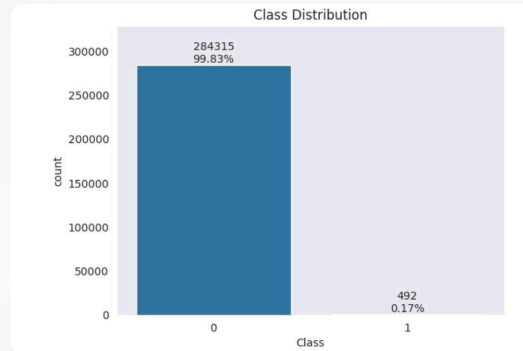
```
df1 = pd.read_csv("creditcard.csv")
df = df1.copy()
df
```

	Time	V1	V2	V3	V4	V5	V6	V7	V8	...	V22	V23	V24	V25	V26	V27	V28	Amount	Class
0	0.000	-1.360	-0.073	2.536	1.378	-0.338	0.462	0.240	0.099	...	0.278	-0.110	0.067	0.129	-0.189	0.134	-0.021	149.620	0
1	0.000	1.192	0.266	0.168	0.448	0.060	-0.082	-0.079	0.085	...	-0.639	0.101	-0.340	0.167	0.126	-0.009	0.015	2.690	0
2	1.000	-1.358	-1.340	1.773	0.380	-0.503	1.800	0.791	0.248	...	0.772	0.909	-0.689	-0.328	-0.139	-0.055	-0.060	378.690	0
3	1.000	-0.966	-0.185	1.793	-0.863	-0.010	1.247	0.238	0.377	...	0.005	-0.190	-1.176	0.647	-0.222	0.063	0.061	123.500	0
4	2.000	-1.158	0.878	1.549	0.403	-0.407	0.096	0.593	-0.271	...	0.798	-0.137	0.141	-0.206	0.502	0.219	0.215	69.990	0
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
284802	172786.000	-11.881	10.072	-9.835	-2.067	-5.364	-2.607	-4.918	7.305	...	0.112	1.014	-0.509	1.437	0.250	0.944	0.824	0.770	0
284803	172787.000	-0.733	-0.055	2.035	-0.739	0.868	1.058	0.024	0.295	...	0.924	0.012	-1.016	-0.607	-0.395	0.068	-0.054	24.790	0
284804	172788.000	1.920	-0.301	-3.250	-0.558	2.631	3.031	-0.297	0.708	...	0.578	-0.038	0.640	0.266	-0.087	0.004	-0.027	67.880	0
284805	172788.000	-0.240	0.530	0.703	0.690	-0.378	0.624	-0.686	0.679	...	0.800	-0.163	0.123	-0.569	0.547	0.109	0.105	10.000	0
284806	172792.000	-0.533	-0.190	0.703	-0.506	-0.013	-0.850	1.577	-0.415	...	0.643	0.377	0.009	-0.474	-0.818	-0.002	0.014	217.000	0

284807 rows x 31 columns

CLASS IMBALANCE

Fraud vs. Genuine Transactions



Transaction Type	Count	Percentage
Genuine	284,315	99.83%
Fraud	492	0.17%

Key Insights

Fraudulent transactions are **extremely rare** at only 0.17% of all records.

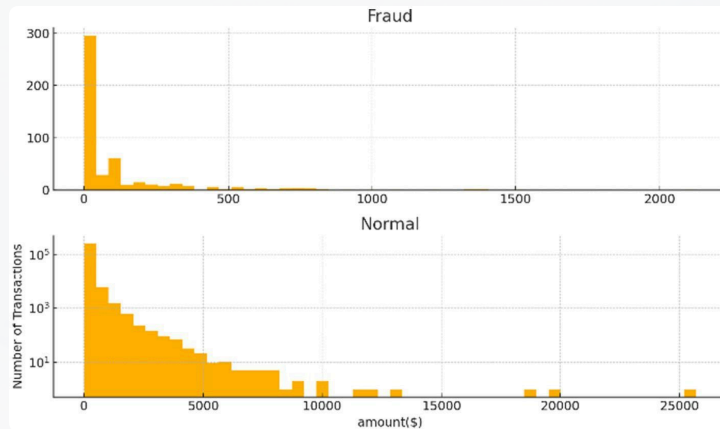
The dataset is **highly imbalanced**, posing a challenge for standard classifiers.

A naive model may become **biased toward genuine** transactions and miss all fraud.

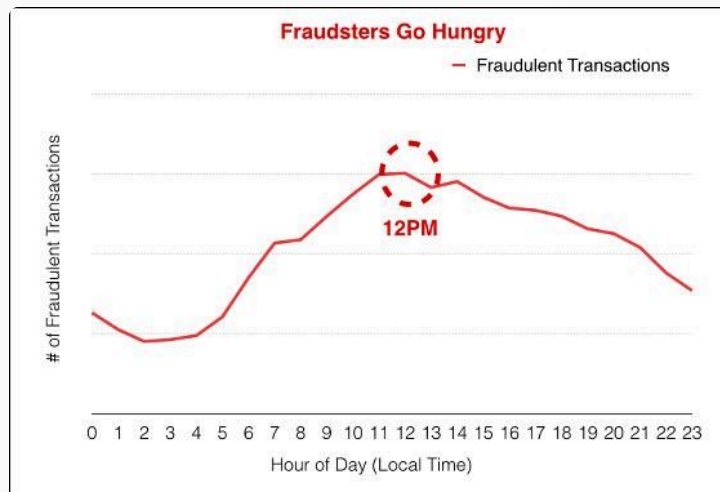
Detecting the minority class is more valuable than achieving high overall accuracy.

Transaction Patterns & Distributions

Transaction Amount Distribution



Time-of-Day Analysis



Key Observations



Amount Ranges

Genuine transactions occur across all amount ranges, while fraud cases concentrate in low and medium-value transactions.



Timing Patterns

Transaction timing reveals different activity patterns — fraudulent transactions peak around 12PM and can help identify suspicious behavior.

EVALUATION STRATEGY

Why Accuracy is Misleading

Example Scenario

Suppose there are **100,000 transactions**: Genuine = 99,830 | Fraud = 170

If the model predicts **every transaction as genuine**:

Accuracy = 99.83% — looks excellent on paper

Fraud Detected = 0 — the model fails completely

Although accuracy is 99.83%, the model fails to detect a single fraudulent transaction.

Better Evaluation Metrics



Precision

Of predicted frauds, how many are real?



Recall

Of all real frauds, how many are caught?



F1 Score

Harmonic mean of Precision and Recall



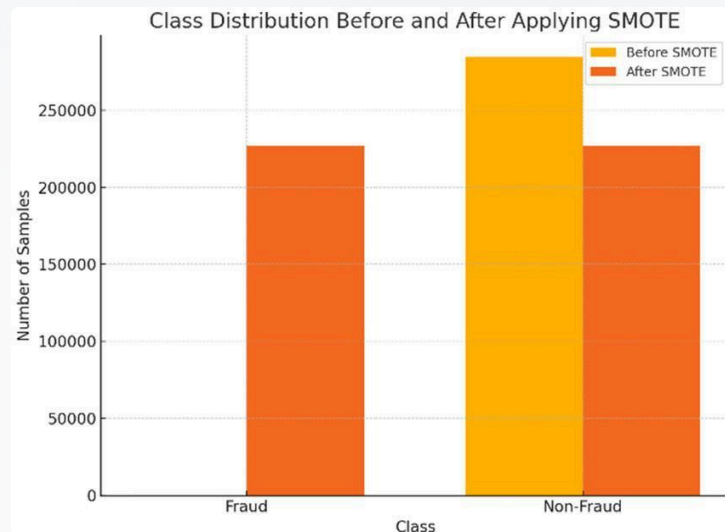
ROC-AUC

Overall discriminative power of the model

In fraud detection, **Recall** is more important than accuracy.

Handling Class Imbalance with SMOTE

SMOTE (Synthetic Minority Oversampling Technique) generates synthetic fraud samples to balance the training dataset.



Preprocessing Steps

01

Train-Test Split

Split data into training and test sets (80/20)

03

Balance Classes

Balance fraud and genuine classes in training data

Why Apply SMOTE?

Better Learning

Improves learning of rare fraud patterns

No Overfitting

Avoids overfitting to the genuine majority

02

Apply SMOTE

Apply SMOTE only to the training data, not test set

04

Train Models

Train machine learning models on balanced data

Reduces Bias

Prevents model bias toward majority class

Machine Learning Pipeline

Train-Test Split

Training Set: 80% of data
Testing Set: 20% of data
Stratified Sampling: Preserves fraud ratio in both sets

Models Used

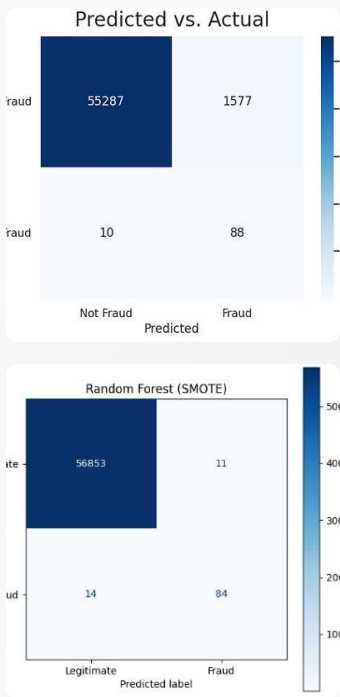
Model	Purpose
Logistic Regression	Baseline classification model
Random Forest	Ensemble model for better fraud detection

ML Pipeline Steps



Confusion Matrices & ROC-AUC

Confusion Matrices



Performance Comparison

Metric	Log. Regression	Random Forest
Accuracy	High	Higher
Precision	Good	Excellent
Recall	Moderate	Highest
F1 Score	Good	Best

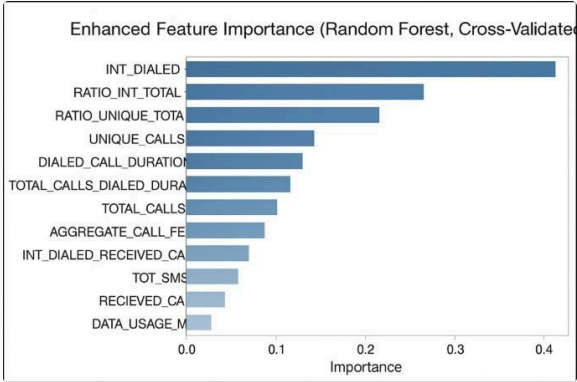
Random Forest detects more fraudulent transactions while maintaining strong overall performance.

Logistic Regression AUC = 0.97 | Random Forest AUC = 0.98

CONCLUSION

Feature Importance & Final Recommendations

Random Forest Feature Importance



Most Influential Features

- 1 V14
- 2 V12
- 3 V10
- 4 Amount
- 5 Time

Business Recommendations



Deploy Random Forest

Use Random Forest for production due to highest Recall, F1, and ROC-AUC.



Prioritize Recall

Optimize for Recall over raw accuracy to minimize missed frauds.



Use SMOTE in Training

Always apply SMOTE during training to improve fraud detection of minority class.



Monitor & Retrain

Continuously monitor performance and retrain with emerging fraud patterns.

This project successfully addressed class imbalance and demonstrated that **Random Forest** is the most suitable model for real-world fraud detection.